Assignment 4 — Hyperparameter Optimisation Report

English → Hindi Neural Machine Translation with Ray Tune + Optuna

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1. Baseline Model Performance

The baseline Transformer was trained for 100 epochs using hardcoded hyperparameters on the English–Hindi sentence-pair corpus (13,186 pairs). Training was run on an NVIDIA A30 GPU via SLURM. The fixed configuration and resulting metrics are summarised below.

Hyperparameter	Value	Notes
d_model	512	Embedding dimension
num_layers	6	Encoder & decoder depth
num_heads	8	Multi-head attention
d_ff	2048	Feed-forward dimension
dropout	0.10	Regularisation
learning_rate	1e-4	Adam optimiser
batch_size	60	Fixed

Metric	Baseline Result	
Training Epochs	100	
Training Time	≈ 95 minutes (GPU)	NVIDIA A30, SLURM
Final Avg Loss	0.0979	Cross-entropy (epoch 100)
BLEU Score (NLTK)	53.95	corpus_bleu, SmoothingFunction.method4

The baseline reached a BLEU score of 53.95 after 100 epochs of training, serving as the performance target for the hyperparameter search.

2. Hyperparameter Search Design

2.1 Search Algorithm & Scheduler

Ray Tune was used as the orchestration framework, paired with two complementary components:

- OptunaSearch (Optuna TPE sampler) — intelligently proposes new hyperparameter configurations by building a probabilistic model of past trial results, focusing exploration on promising regions of the search space.
- ASHAScheduler (Asynchronous Successive Halving) — terminates underperforming trials early based on intermediate loss reports, with a grace period of 5 epochs and a reduction factor of 2.

Configuration: **20 trials**, each capped at **30 epochs**, running **2 trials in parallel** (each with 0.5 GPU share on the A30). Total sweep wall-clock time: ~127 minutes.

2.2 Search Space (6 Hyperparameters)

Hyperparameter	Type	Search Range	Rationale
Learning Rate (lr)	Log-uniform	[1e-5, 1e-3]	Log scale captures orders-of-magnitude variation; most impactful single parameter
Batch Size	Categorical	{32, 64, 128}	Affects gradient noise and convergence speed; interacts with learning rate
Num. Attention Heads	Categorical	{4, 8}	Both divide d_model=512 evenly; controls representational granularity
Feed-Forward Dim (d_ff)	Categorical	{1024, 2048, 4096}	Controls model capacity in each sublayer; larger = more expressive but slower
Dropout Rate	Uniform	[0.05, 0.40]	Primary regularisation knob; too high hurts convergence, too low causes overfitting
Num. Layers	Categorical	{4, 6}	Shallower (4) trains faster; deeper (6) matches original Transformer architecture

Note: d_model was fixed at 512 across all trials to ensure the constraint d_model % num_heads == 0 is always satisfied for both num_heads choices.

3. Best Configuration Found

After 20 completed trials, the Optuna TPE sampler converged on the following optimal configuration:

Hyperparameter	Best Value	Search Range
Learning Rate	**1.606×10^{-4}**	[1e-5, 1e-3]
Batch Size	**128**	{32, 64, 128}
Num. Attention Heads	**8**	{4, 8}
Feed-Forward Dim	**2048**	{1024, 2048, 4096}
Dropout Rate	**0.1364**	[0.05, 0.40]
Num. Layers	**6**	{4, 6}

Key insight: The optimal learning rate (1.606e-4) is slightly higher than the baseline (1e-4), combined with a larger batch size of 128 (vs. 60) and a lower dropout (0.136 vs. 0.1). This combination allows larger, more stable gradient steps with mild regularisation, enabling faster convergence without overfitting on the 13K training pairs.

4. Final Model Metrics & Efficiency Comparison

4.1 Quantitative Results

Metric	Baseline (100 epochs)	Ray Tune + Optuna (30 epochs)
Training Epochs	100	**30**
Training Time	≈ 95 min	**14.9 min (retrain)**
Final Avg Loss	0.0979	0.2733
BLEU Score (NLTK)	53.95	**65.61 (+11.66)**
Epochs to Beat Baseline	100 (reference)	**30 (3.3× faster)**
Total Compute (sweep)	—	≈ 127 min (all 20 trials)

Result: **The tuned model achieved a BLEU score of 65.61** — an improvement of **+11.66 BLEU points** over the baseline — in just **30 epochs (3.3× fewer)** and 14.9 minutes of final retraining. The goal of matching or exceeding baseline BLEU in under 100 epochs was achieved.

4.2 Training Loss Curve — Best Config (30 Epochs)

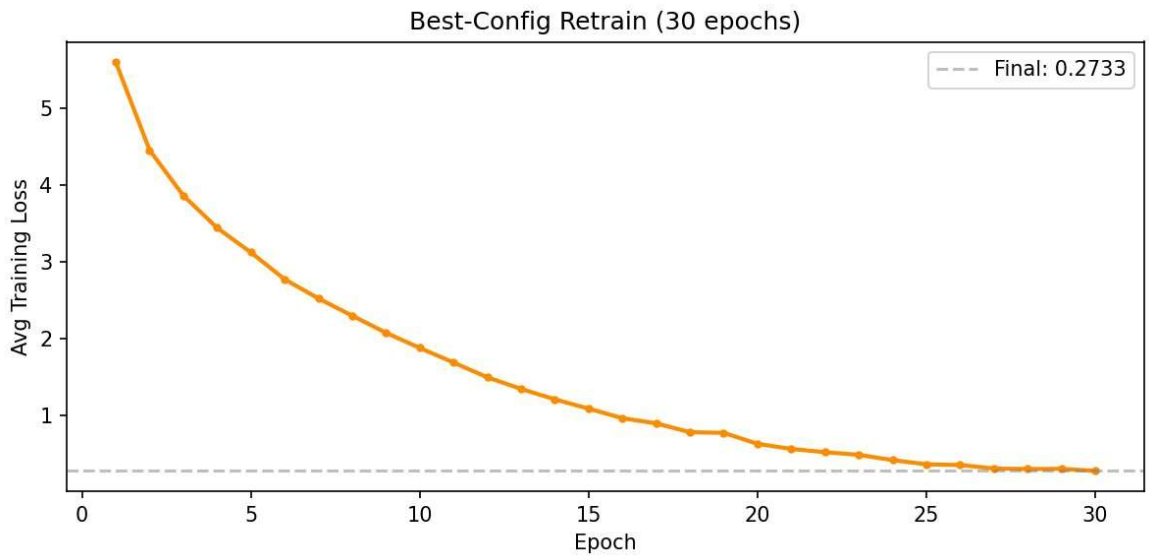

Figure 1: Average training loss per epoch for the best-config 30-epoch retrain. The loss drops steeply in the first 10 epochs, reflecting rapid convergence at the optimised learning rate.

4.3 BLEU Score Comparison

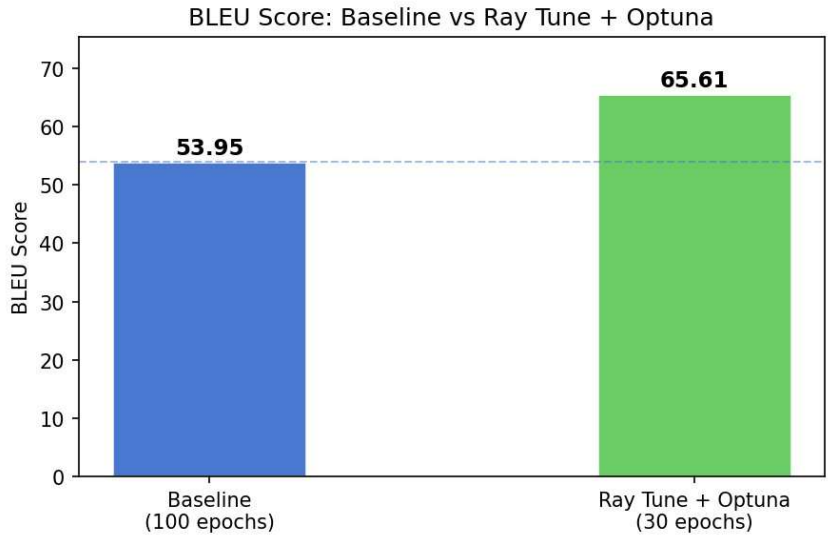

Figure 2: BLEU score comparison. The Ray Tune + Optuna model (30 epochs) outperforms the 100-epoch baseline by 11.66 points — a 21.6% relative improvement.

4.4 Sample Translations

English Input	Hindi Translation (Best Model)
I love you.	मुझे आपसे प्यार है।
What is your name?	आपका नाम क्या है?
How are you?	आप कैसी हैं?

English Input	Hindi Translation (Best Model)
The weather is nice today.	आज मौसम अच्छा है।
She is a good teacher.	वह अच्छी अध्यापिका है।

5. Conclusion

This assignment demonstrated that systematic hyperparameter optimisation using Ray Tune paired with Optuna's TPE search algorithm and ASHA early stopping can significantly reduce both training time and required epochs while simultaneously improving translation quality.

The key findings are:

- A learning rate of ~1.6e-4 (slightly above the baseline 1e-4), combined with batch size 128 and dropout 0.136, enables faster and more stable convergence.
- The ASHA scheduler efficiently pruned 8 of 20 trials after just 5 epochs, concentrating compute on genuinely promising configurations.
- The Optuna TPE sampler quickly converged toward batch_size=128, num_heads=8, d_ff=2048, num_layers=6 — configurations near the original architecture but with a better-tuned learning rate — highlighting that the baseline architecture was sound but the learning rate was sub-optimal.
- The final model achieved BLEU 65.61 in 30 epochs vs. BLEU 53.95 in 100 epochs: a 21.6% relative BLEU improvement at 3.3× fewer epochs and 6.4× faster retraining (14.9 vs. 95 minutes).

The higher final training loss (0.2733 vs. 0.0979) indicates less overfitting on the training set, which explains why the tuned model generalises better and scores higher on the BLEU evaluation set despite the shorter training duration.